THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapters 1–4: **Chemical Reactions and Equations, Acids, Bases and Salts, Metals and Non-metals, Carbon and its Compounds** | Class 10 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(d)** one or more new substances	Chemical changes form new substances.
2	**(b)** sour	Acids taste sour.
3	**(b)** malleability	Malleability.
4	**(d)** 6	Carbon has 6 protons.
5	**(a)** combination reaction	Two reactants form one product.
6	**(a)** 7	Neutral solutions have pH 7.
7	**(b)** gold	Gold is at the bottom of the series.
8	**(a)** ethane	Alkane.
9	**(a)** blue	Hydrated copper(II) ions are blue.
10	**(b)** salt and water	Neutralisation.
11	**(b)** haematite	Haematite is Fe_2O_3.
12	**(b)** allotropes of carbon	Both are pure carbon.
13	**(d)** reactants	Reactants are written on the left.
14	**(b)** soapy	Bases are soapy to touch.
15	**(a)** ductility	Ductility.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Mass can neither be created nor destroyed in a chemical reaction (1); Total mass of reactants equals total mass of products (1).	1+1
17	Substance giving H^+ ions in water (1); HCl, H_2SO_4 (1).	1+1
18	Malleability: beaten into sheets (1); Ductility: drawn into wires (1).	1+1
19	Valency is 4 (1); It has 4 electrons in its outer shell (1).	1+1
20	A single compound splits into two or more simpler substances (1); $CaCO_3 \longrightarrow CaO + CO_2$ (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Count atoms: 1 Na, 2 Cl on the left, 1 Na, 1 Cl on the right (1); Put 2 before NaCl and 2 before Na (1); $2\,Na + Cl_2 \longrightarrow 2\,NaCl$ (1).	1+1+1
22	Lemon juice: blue litmus turns red (1); Soap solution: red litmus turns blue (1); Phenolphthalein: colourless in lemon juice, pink in soap (1).	1+1+1
23	Lustre (1); Malleable and ductile (1); Good conductors of heat and electricity (1).	1+1+1
24	Methane CH_4 (1); Ethane C_2H_6 (1); Propane C_3H_8 (1).	1+1+1
	OR: Combination (1); Two reactants form a single product (1); Exothermic (heat is released) (1).	1+1+1
25	Sour taste (1); Turn blue litmus red (1); React with metals to give hydrogen (1).	1+1+1
	OR: $2\,Na + 2\,H_2O \longrightarrow 2\,NaOH + H_2$ (2); Sodium hydroxide and hydrogen (1).	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Burns with a dazzling white flame (1); $2\,Mg + O_2 \longrightarrow 2\,MgO$ (1); Combination reaction (1); Product is magnesium oxide (1); To remove the oxide layer (1).	1+1+1+1+1
	OR: Acid sour, base bitter (1); Acid turns blue litmus red, base turns red litmus blue (1); Acid gives H^+, base gives OH^- (1); Acid pH below 7, base pH above 7 (1); Examples: HCl, NaOH (1).	1+1+1+1+1

27	Lustre: metals lustrous, non-metals dull (1); Conduction: metals conduct (1); Malleability: metals malleable, non-metals brittle (1); Hardness: metals generally hard (1); Oxides: basic vs acidic (1).	1+1+1+1+1
	OR: CH_4: 4 C-H single bonds (1); C_2H_4: 4 C-H and a C=C (1); H_2O: 2 O-H (1); Shared pairs: 4, 6 and 2 (1); Covalent bonds form by sharing (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapters 1–4: **Chemical Reactions and Equations, Acids, Bases and Salts, Metals and Non-metals, Carbon and its Compounds** | Class 10 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** 4	Tetravalent.
2	**(d)** decomposition reaction	One reactant gives two products.
3	**(b)** acidic	pH below 7 is acidic.
4	**(a)** potassium	K is at the top.
5	**(d)** ethene	Alkene.
6	**(c)** white	Barium sulphate is a white insoluble salt.
7	**(b)** hydrogen gas	Hydrogen is released.
8	**(d)** bauxite	Bauxite.
9	**(a)** graphite	Graphite has free electrons.
10	**(a)** CaO	Quick lime is calcium oxide.
11	**(a)** acid	Acids turn blue litmus red.
12	**(d)** lustre	Metallic lustre.
13	**(b)** methane	$\mathrm{CH_4}$.
14	**(a)** displacement reaction	Fe displaces Cu.
15	**(c)** basic	pH above 7 is basic.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Colourless in acid (1); Pink in base (1).	1+1
17	Brass: Cu and Zn (1); Bronze: Cu and Sn (1).	1+1
18	Same molecular formula (1); Different structures (1).	1+1
19	Slow destruction of a metal by air, moisture or acids (1); Rust ($\mathrm{Fe_2O_3 \cdot \mathit{x}\, H_2O}$) (1).	1+1
20	Acid + base (1); gives salt + water (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Low melting and boiling points (1); Poor conductors of electricity (1); Mostly burn in air (1).	1+1+1
22	Nail becomes covered with a reddish-brown deposit (1); Blue colour of the solution fades (1); Solution turns pale green (1).	1+1+1
23	pH below 5.6 is acid rain (1); It makes river water acidic (1); Survival of aquatic life becomes difficult (1).	1+1+1
24	Copper: wires (1); Aluminium: foil (1); Iron: bridges and machines (1).	1+1+1
	OR: Sodium ethanoate and water (1); $\mathrm{CH_3COOH + NaOH \longrightarrow CH_3COONa + H_2O}$ (2).	1+2
25	Left: 2 N, 2 H; right: 1 N, 3 H (1); Put 2 before NH3 and 3 before H2 (1); $\mathrm{N_2 + 3\,H_2 \longrightarrow 2\,NH_3}$ (1).	1+1+1
	OR: Magnesium chloride and hydrogen (1); $\mathrm{Mg + 2\,HCl \longrightarrow MgCl_2 + H_2}$ (1); Hydrogen burns with a pop sound (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Slow attack on a metal by moisture and air (1); Needs air (oxygen) and moisture (1); Rust is $\mathrm{Fe_2O_3 \cdot \mathit{x}\, H_2O}$ (1); Prevention: painting, oiling (1); Galvanisation, alloying (1).	1+1+1+1+1
	OR: $\mathrm{Na_2CO_3 \cdot 10\,H_2O}$; prepared from baking soda (1); Uses: glass, soap, hard water (1); $\mathrm{CaSO_4 \cdot \tfrac{1}{2}\,H_2O}$; heat gypsum at 373 K (1); $\mathrm{CaSO_4 \cdot 2\,H_2O \longrightarrow CaSO_4 \cdot \tfrac{1}{2}\,H_2O + 1.5\,H_2O}$ (1); Uses: bone plaster, toys (1).	1+1+1+1+1
27	Brass: Cu + Zn (1); Bronze: Cu + Sn (1); Solder: Pb + Sn (1); Stainless steel: Fe, Ni, Cr (1); Amalgam: Hg alloy (1).	1+1+1+1+1

	OR: $CH_3COOH + NaOH \longrightarrow CH_3COONa + H_2O$ (1); $2\,CH_3COOH + Na_2CO_3 \longrightarrow 2\,CH_3COONa + H_2O + CO_2$ (1); $CH_3COOH + C_2H_5OH \longrightarrow CH_3COOC_2H_5 + H_2O$ (1); Esterification needs conc. H_2SO_4 (1); Ester smells fruity (1).	1+1+1+1+1

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ANSWER KEY & MARKING SCHEME SET 3

Chapters 1–4: **Chemical Reactions and Equations, Acids, Bases and Salts, Metals and Non-metals, Carbon and its Compounds** | Class 10 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** iron	Iron rusts.
2	**(a)** ethyne	Alkyne.
3	**(a)** yellow	Golden-yellow solid.
4	**(c)** carbon dioxide	CO_2 is evolved.
5	**(c)** copper glance	Cu_2S.
6	**(a)** diamond	Diamond.
7	**(d)** $Ca(OH)_2$	Slaked lime is calcium hydroxide.
8	**(b)** base	Bases turn red litmus blue.
9	**(c)** sonorous	Sonorous.
10	**(c)** CH_4	One C, four H.
11	**(a)** double displacement reaction	Exchange of ions.
12	**(a)** the approximate pH	It shows different colours for different pH.
13	**(c)** $Fe_2O_3 \cdot x\,H_2O$	Hydrated iron oxide.
14	**(b)** saturated hydrocarbons	Single bonds only.
15	**(a)** reddish-brown	Copper metal is reddish brown.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Magnesium oxide (1); Equation $2\,Mg + O_2 \longrightarrow 2\,MgO$ (1).	1+1
17	Alkane $CnH_{2n}n_2^+$ (1); Alkene $CnH_{2}n$ (1).	1+1
18	Magnesium + oxygen (1); gives magnesium oxide (1).	1+1
19	Substance that gives OH^- ions in water / reacts with acids (1); NaOH, KOH (1).	1+1
20	Silver (1); Copper (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Dull appearance (1); Brittle and non-conductors (1); Low melting points (1).	1+1+1
22	Ethene C_2H_4 (1); Propene C_3H_6 (1); Butene C_4H_8 (1).	1+1+1
23	Decomposition (1); One compound gives two products (1); Needs heat, so thermal decomposition (1).	1+1+1
24	Bitter taste and soapy touch (1); Turn red litmus blue (1); React with acids to give salt and water (1).	1+1+1
	OR: $MgO + H_2O \longrightarrow Mg(OH)_2$ (2); Magnesium hydroxide is basic (1).	2+1
25	Solvent (1); Fuel mixed with petrol (1); In medicines and tinctures (1).	1+1+1
	OR: Burns with a dazzling white flame (1); White powder is formed (1); Heat and light are given out (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Brown fumes are seen (1); $2\,Pb(NO_3)_2 \longrightarrow 2\,PbO + 4\,NO_2 + O_2$ (1); Thermal decomposition (1); Brown gas is nitrogen dioxide (1); Residue is yellow lead oxide (1).	1+1+1+1+1
	OR: $Zn + 2\,HCl \longrightarrow ZnCl_2 + H_2$ (1); $NaOH + HCl \longrightarrow NaCl + H_2O$ (1); $Na_2CO_3 + 2\,HCl \longrightarrow 2\,NaCl + H_2O + CO_2$ (1); $CuO + 2\,HCl \longrightarrow CuCl_2 + H_2O$ (1); $NaHCO_3 + HCl \longrightarrow NaCl + H_2O + CO_2$ (1).	1+1+1+1+1
27	$2\,Na + 2\,H_2O \longrightarrow 2\,NaOH + H_2$ (1); Na reacts violently (1); $Mg + 2\,H_2O \longrightarrow Mg(OH)_2 + H_2$ (hot water) (1); $3\,Fe + 4\,H_2O \longrightarrow Fe_3O_4 + 4\,H_2$ (steam) (1); Hydrogen is released (1).	1+1+1+1+1

	OR: Single bonds vs double/triple bonds (1); General formula differs (1); Saturated: clean flame (1); Unsaturated: sooty flame (1); Addition reactions in unsaturated (1).	1+1+1+1+1

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